SG No. 105 C_{4v}^{7} $P4_2mc$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `2mm.`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,5,6]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3,4,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[3,4,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[3,4,7,8]`

Table 4: Wyckoff site: `4d`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,5]`
2	$[-x,\ 0,\ z]$	`[2,6]`
3	$\left[0,\ x,\ z+\frac{1}{2}\right]$	`[3,8]`
4	$\left[0,\ -x,\ z+\frac{1}{2}\right]$	`[4,7]`

Table 5: Wyckoff site: `4e`, site symmetry: `.m.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	`[1,5]`
2	$\left[-x,\ \frac{1}{2},\ z\right]$	`[2,6]`
3	$\left[\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[3,8]`
4	$\left[\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	`[4,7]`

Table 6: Wyckoff site: `8f`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	`[4]`

continued ...

Table 6

No.	position	mapping
5	$[x, -y, z]$	[5]
6	$[-x, y, z]$	[6]
7	$\left[-y, -x, z+\frac{1}{2}\right]$	[7]
8	$\left[y, x, z+\frac{1}{2}\right]$	[8]